INHOLLAND
Domein Techniek, Ontwerpen en Informatica | Cluster ICT

Calculus voor technisch Informatici

Lecture
Piecewise interpolation

Piecewise interpolation

How to define piecewise functions that interpolate measurement values ?

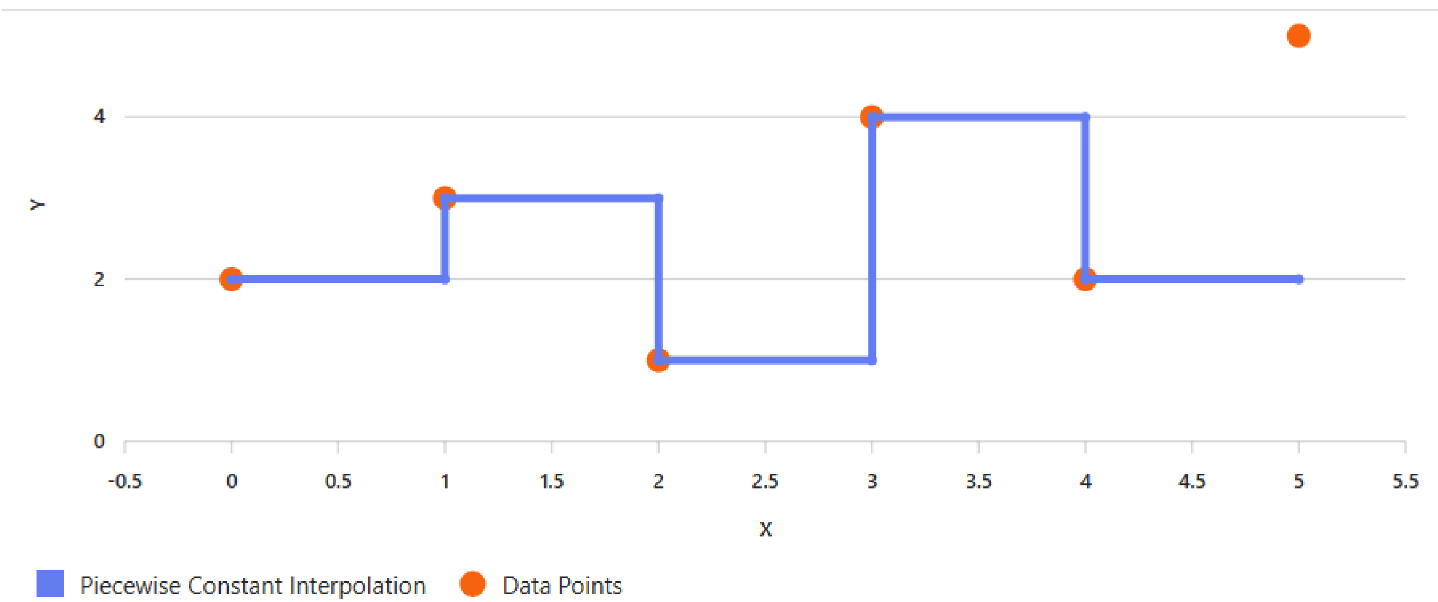
Yes of course we wil use polynomials to interpolate measurement values.

$$p(x) = \ a_n x^n + a_{n-1}\, x^{n-1} + \cdots + a_1\, x^1 + a_0$$

With this we can “approximate real situation”

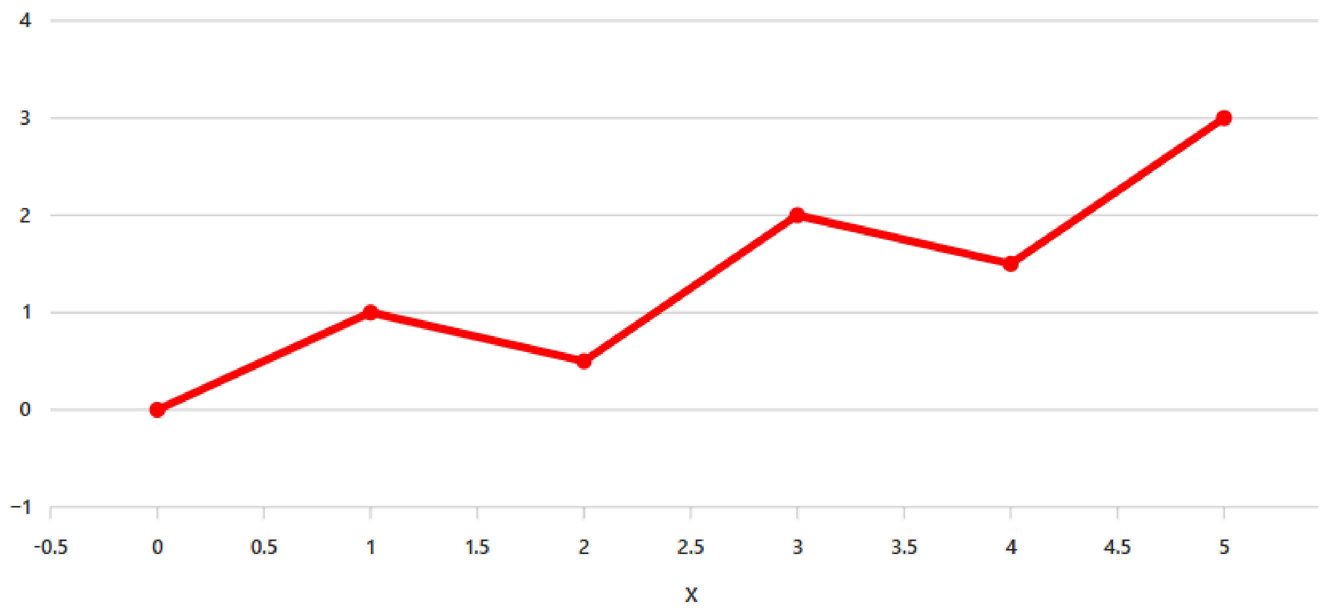
Piecewise interpolation

Piecewise constant function



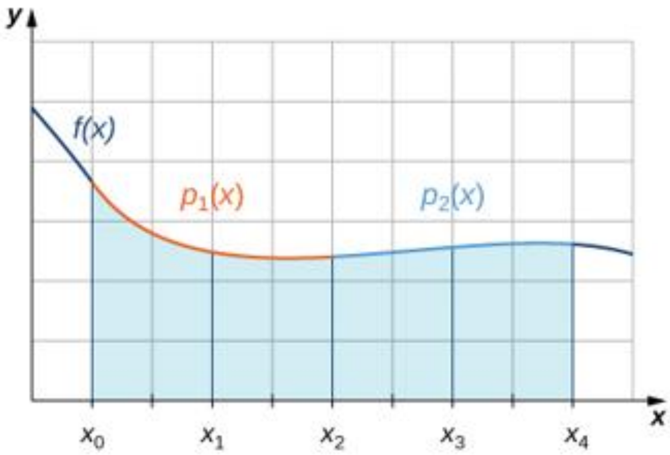
Piecewise interpolation

Piecewise linear function



Piecewise interpolation

Piecewise quadratic function



Piecewise interpolation - Lagrange

How to define the functions for the intervals?

Constant functions are easy !

$$x \epsilon [x_i, x_{i+1}], p_0(x) = f(x_i)$$

Linear function are also more complex

$x \epsilon [x_i, x_{i+1}]$,

$$p_1(x) = \frac{x - x_{i+1}}{x_i - x_{i+1}} f(x_i) + \frac{x - x_i}{x_{i+1} - x_i} f(x_{i+1})$$

Piecewise interpolation

Finally the quadratic ones
$x\epsilon[x_{i-1}, x_i, x_{i+2}]$,

$p_2(x) =$
$\frac{(x-x_i)(x-x_{i+1})}{(x_{i-1}-x_i)(x_{i-1}-x_{i+1})} f(x_{i-1})$

$+ \frac{(x-x_{i-1})(x-x_{i+1})}{(x_i-x_{i-1})(x_i-x_{i+1})} f(x_i)$

$+ \frac{(x-x_{i-1})(x-x_i)}{(x_{i+1}-x_{i-1})(x_{i+1}-x_i)} f(x_{i+1})$

Knowing this we can easily define piecewise functions …..

By the way these types of polynomials are called Lagrange polynoms

Joseph-Louis Lagrange

Wiki:

Joseph-Louis Lagrange (born **Giuseppe Luigi Lagrangia** or **Giuseppe Ludovico De la Grange Tournier**; 25 January 1736 – 10 April 1813), also reported as **Giuseppe Luigi Lagrange** or **Lagrangia**, was an Italian and naturalized French mathematician, physicist and astronomer. He made significant contributions to the fields of analysis, number theory, and both classical and celestial mechanics.